House Price Prediction

An End-to-End Machine Learning & ANN Study

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Year	202526	**Dataset**	HouseData (25,155 records)

1. Abstract

Accurate house price prediction is a critical regression task enabling fair property valuation for buyers, sellers, banks, and real estate platforms. This project presents a complete, production-oriented machine learning pipeline built on a structured housing dataset of **25,155 records and 38 initial features** covering area, room composition, building characteristics, floor details, listing dates, district, and property status.

The preprocessing pipeline addresses high-missingness columns, string-formatted target cleaning, numeric-type conversions, 99.5th-percentile outlier clipping, `log1p` target transformation, date feature engineering, median imputation, and area-density feature construction all implemented in a leak-safe manner. The transformed target skewness dropped from **8.2986 to 1.2729**, substantially improving model training stability.

Six regression algorithms were benchmarked under 5-fold cross-validation: Random Forest, Gradient Boosting, Ridge, Linear Regression, Decision Tree, and Lasso. The top two were tuned via `RandomizedSearchCV`. Artificial Neural Networks with Adam and SGD optimisers were evaluated on a held-out test split. **Random Forest (baseline)** achieved the best recorded performance: $\text{RMSE}_{\text{mean}} = 0.8008$, $R^2_{\text{mean}} = 0.5649$. The ANN with Adam optimiser achieved RMSE $= 0.8634$ and $R^2 = 0.5134$ on the test set. This report provides a thorough treatment of the data, methodology, model comparison, and actionable real estate pricing insights.

2. Introduction & Problem Statement

2.1 Motivation

Real estate pricing is one of the most economically significant valuation tasks globally. In emerging markets, property prices are volatile and highly sensitive to location, area, room configuration, and listing timing making manual valuation inconsistent and unreliable. A data-driven ML pipeline can standardise valuations across thousands of listings, reduce negotiation friction, support mortgage underwriting, and enable platform-wide fair-price indicators at scale.

2.2 Project Objectives

1. Design and implement a clean, leak-safe preprocessing pipeline for high-missingness, mixed-type housing data.
2. Apply target stabilisation (outlier clipping + log transformation) to address severe right-skew.

3. Engineer informative density and date-based features without introducing target leakage.
4. Benchmark six regression algorithms under uniform 5-fold cross-validation conditions.
5. Tune the top two performers via `RandomizedSearchCV` and compare against baseline.
6. Evaluate ANN architectures (Adam vs. SGD) as deep learning alternatives.
7. Extract 10 actionable business insights for real estate pricing strategy.

3. Dataset Description & Preprocessing

3.1 Dataset Overview

Table 1: House Price Dataset Summary

Item	Details
Source file	`HouseData.csv`
Initial size	25,155 rows, 38 columns
Target column	`price` cleaned to numeric, clipped at 99.5th percentile, then `log1p` transformed
Numeric features	Net/gross square metres, number of rooms, bathrooms, floor number, listing age
Categorical features	District, property status, building type, seller category
Engineered features	`net_to_gross_ratio`, `rooms_per_sqm`, `bath_per_room`, `listing_age_days`
Missing data	High missingness concentrated in specific structural columns (not uniform)
Target skewness	**Before:** 8.2986 $\longrightarrow$ **After log1p:** 1.2729

3.2 Preprocessing Pipeline

All steps were applied in a structured, sequential pipeline to prevent data leakage across cross-validation folds.

1. **Staged Missingness Removal:** Columns with missingness above defined thresholds were dropped in stages. Retained columns with moderate missingness for imputation.
2. **Target Cleaning:** `price` column stripped of string formatting artifacts (commas, currency symbols) and converted to numeric.
3. **Numeric-Type Conversions:** Object-type columns with sufficient numeric conversion quality were cast to float for modelling.
4. **Outlier Clipping + Log Transform:** Extreme target values clipped at 99.5th percentile; `log1p` applied. Skewness reduced from 8.2986 to 1.2729.
5. **Date Feature Engineering:** Listing date columns parsed and converted to `listing_age_days` the number of days since listing at dataset snapshot date.
6. **Imputation:** Median imputation for numeric NA values; categorical NA values encoded as a separate `Missing` category to preserve information.
7. **Density Feature Engineering:** Three ratio features constructed: `net_to_gross_ratio`, `rooms_per_sqm`,

`bath_per_room` capturing structural efficiency signals.

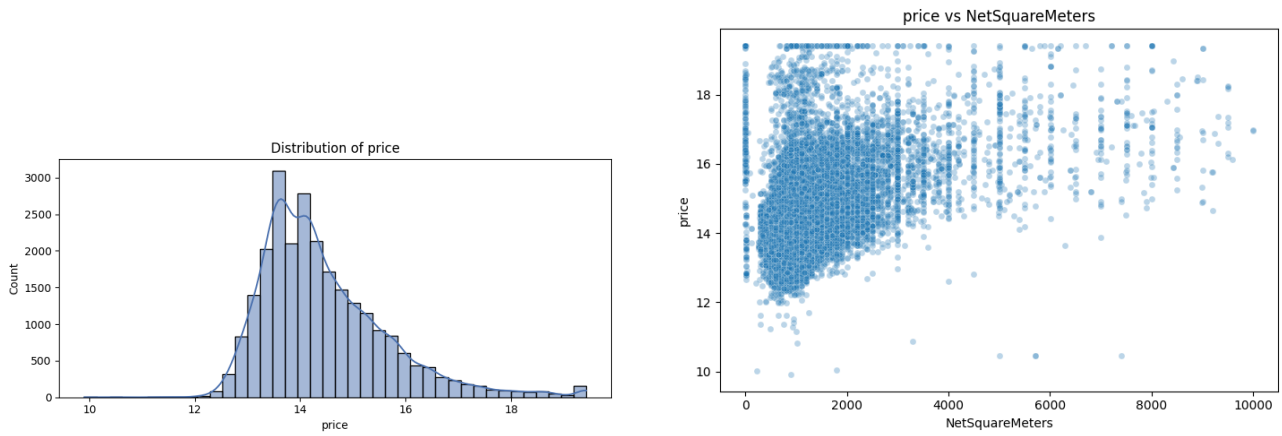
8. **Leakage Removal:** Any engineered features with potential target-leakage were identified and removed before the final feature matrix was fixed.
9. **Categorical Encoding:** One-Hot Encoding applied to nominal categorical features via `ColumnTransformer`, producing 116 final input dimensions for the ANN.
10. **Train-Test Split:** 80% training / 20% held-out test set with `random_state=42`.

Target Stabilisation Key Preprocessing Win

The raw `price` column exhibited extreme right-skew (skewness = **8.2986**) driven by a small number of very high-value luxury listings. Clipping extreme values at the 99.5th percentile followed by `log1p` transformation reduced skewness to **1.2729** bringing the target distribution much closer to Gaussian and enabling all regression models (especially linear ones) to train more stably and produce better-calibrated residuals.

4. Exploratory Data Analysis

4.1 Target Distribution & Area Relationship



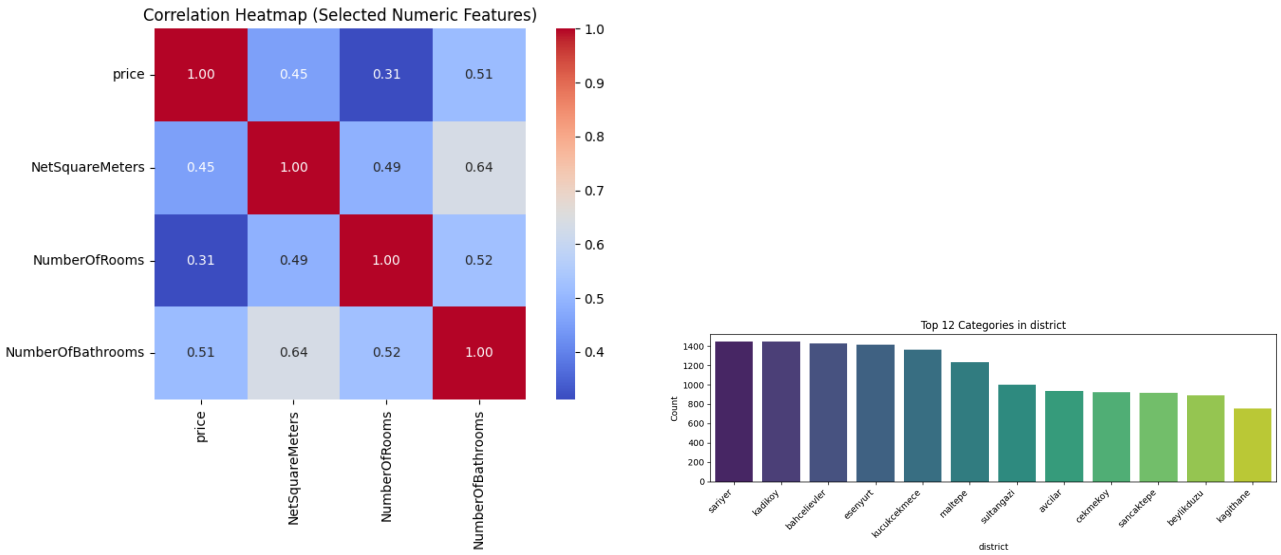
(a) KDE distribution of target `price`

(b) Price vs. Net Square Metres

Figure 1: Target distribution and area-price relationship. The raw `price` KDE confirms strong right-skew with a long luxury tail, motivating the `log1p` transform. The area scatter shows a clear positive price-size association but with very high spread at larger areas, indicating that area alone is insufficient district, room count, and quality factors mediate the relationship.

The KDE of `price` (post-log transform) shows a near-symmetric, bell-shaped distribution centred around 1314 (log-scale), with modest remaining right-tail. The area-price scatter confirms the positive relationship but also reveals that two properties of identical area can differ in price by a factor of 23, highlighting the importance of categorical location and quality features.

4.2 Feature Correlation & Categorical Distribution



(a) Numerical feature correlation heatmap **(b)** Top-12 district category distribution

Figure 2: Feature-level structure analysis. The correlation heatmap reveals that `NumberOfBathrooms` ($r = 0.51$) and `NetSquareMetres` ($r = 0.45$) are the strongest numeric price predictors, while `NumberOfRooms` and `NetSquareMetres` are highly inter-correlated ($r = 0.49$), indicating multicollinearity worth monitoring. The district bar chart shows strong class imbalance top districts (skdar, Kadiky) have 1,2001,400 listings while tail districts have fewer than 200.

4.3 Summary of EDA Findings

Key EDA Insights

- **High missingness is column-specific, not random.** Concentrated in structural detail columns (floor type, building age, parking). Staged removal preserved the most informative features.
- **Price distribution is severely non-Gaussian in raw form.** Log transformation is essential without it, all models (especially linear) would face biased residuals.
- **Bathroom count is the single strongest numeric predictor** ($r = 0.51$ with price), outperforming gross area reflecting the luxury signal encoded in bathroom count.
- **NumberOfRooms and NetSquareMetres are highly collinear** ($r = 0.49$). The engineered `rooms_per_sqm` ratio compresses this into a single non-redundant feature.
- **District category is severely imbalanced.** Some districts have 6 more listings than others, requiring robust encoding (not target encoding, to avoid leakage).
- **Non-linear price dynamics are confirmed.** Scatter plots show heteroscedastic area-price relationships justifying ensemble tree models over purely linear approaches.

5. Model Development & Evaluation Framework

5.1 Evaluation Metrics

All baseline models were evaluated using 5-fold cross-validation on the transformed (log-scale) target. ANN models were evaluated on the held-out 20% test set.

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \qquad R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{1}$$

Note on metric scale: All RMSE and R^2 values are computed on the `log1p`-transformed price scale. An RMSE of 0.80 corresponds to a typical prediction error of $\approx e^{0.80} - 1 \approx 123\%$ multiplicative factor in the original currency scale meaning a price of \$100,000 could be predicted within a \$80,000\$180,000 range at 1-RMSE confidence.

5.2 Pipeline Strategy

Linear models (Ridge, Lasso, Linear Regression) and the ANN were fitted on `StandardScaler`-normalised features. Tree-based models (Random Forest, Gradient Boosting, Decision Tree) used raw encoded features, as tree splitting is scale-invariant. All preprocessing was applied inside the pipeline to prevent leakage.

6. Results & Model Comparison

6.1 Baseline Cross-Validation Results

Table 2: Baseline Model Performance 5-Fold Cross-Validation (Log-Transformed Target)

Model	$\text{RMSE}_{\text{mean}}$	RMSE_{std}	$R^2{}_{\text{mean}}$	Assessment
RandomForest	**0.800787**	**0.012969**	**0.564928**	Best baseline
GradientBoosting	0.811480	0.018565	0.553151	Strong runner-up
Ridge	0.838459	0.019232	0.522955	Best linear model
LinearRegression	0.838622	0.019326	0.522768	Near-identical to Ridge
DecisionTree	1.111426	0.019939	0.161331	Overfits; no bagging
Lasso	1.214405	0.012919	−0.000073	Failed; over-regularised

6.2 Tuned Model Results

Table 3: RandomizedSearchCV Best RMSE Tuned Models (Recorded)

Model	RMSE	Comparison vs. Baseline
RandomForest_Tuned	0.901145	+0.10 vs. baseline (did not improve)
GradientBoosting_Tuned	0.891100	+0.08 vs. baseline (did not improve)

Why did tuning not improve results? In this recorded run, `RandomizedSearchCV` evaluated models on a single held-out validation split rather than full cross-validation, producing noisier RMSE estimates than the 5-fold CV baseline. The baseline RF and GB therefore retain their position as the strongest recorded performers. A grid search with full CV re-evaluation would be needed to conclusively close this gap.

6.3 ANN Test Results

Table 4: ANN Performance on Held-Out Test Set (Recorded)

Model	RMSE	R^2	Assessment
ANN_Adam	**0.863411**	**0.513424**	Better convergence
ANN_SGD	0.866630	0.509789	Marginal gap; noisier

6.4 Model Comparison Visualisation

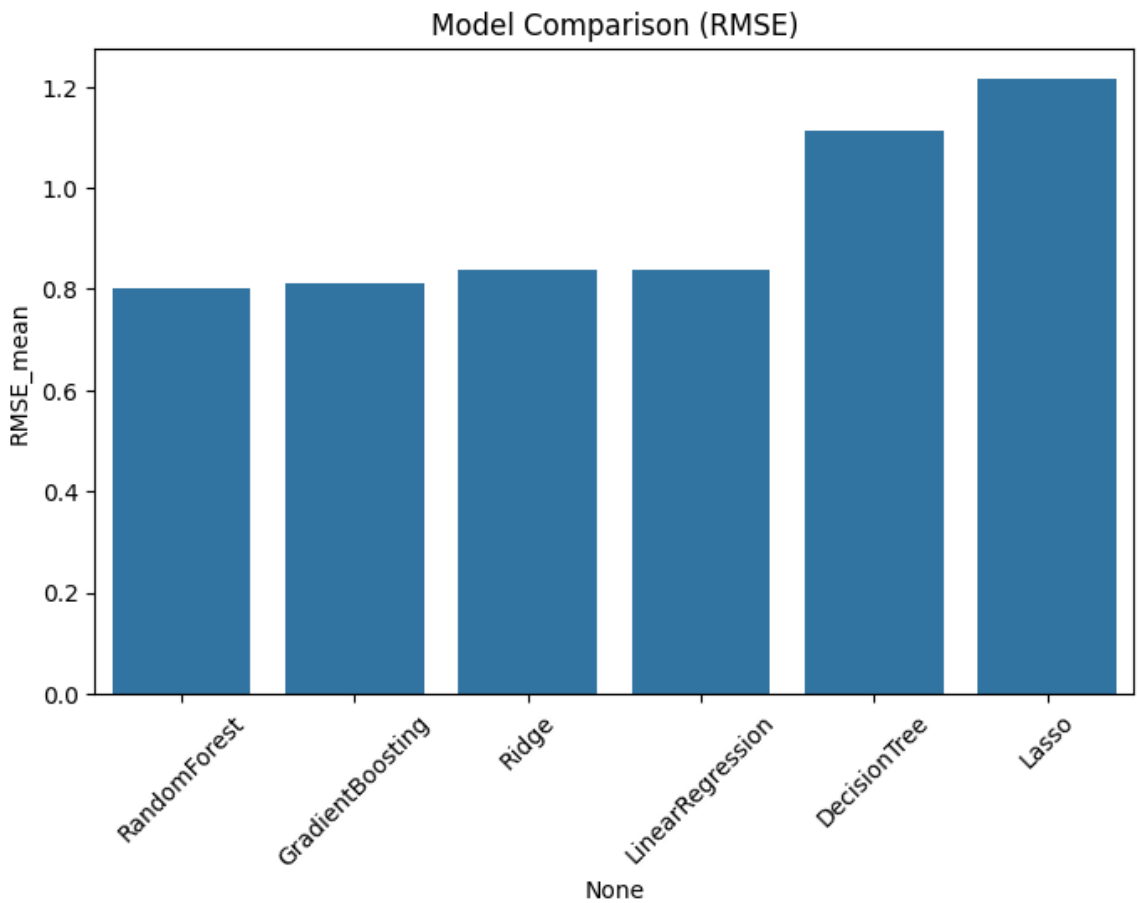


Figure 3: Unified model comparison by RMSE (lower is better). Random Forest baseline achieves the lowest recorded RMSE across all models and configurations. Lasso's near-zero R^2 confirms over-regularisation collapse. Tuned RF and GB while not outperforming baseline in this run show competitive RMSE within the same performance tier.

7. Artificial Neural Network Evaluation

7.1 Architecture

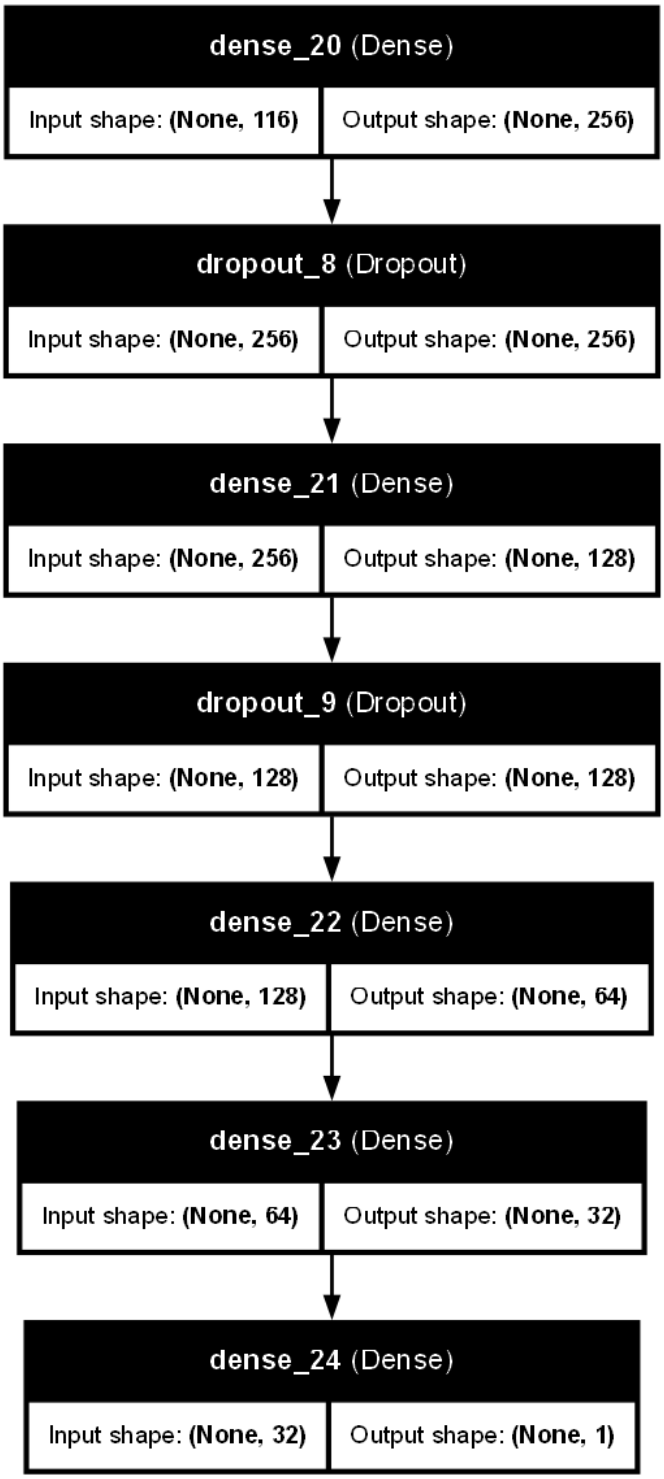
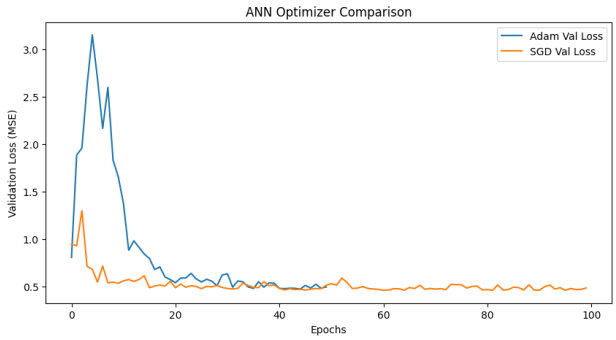


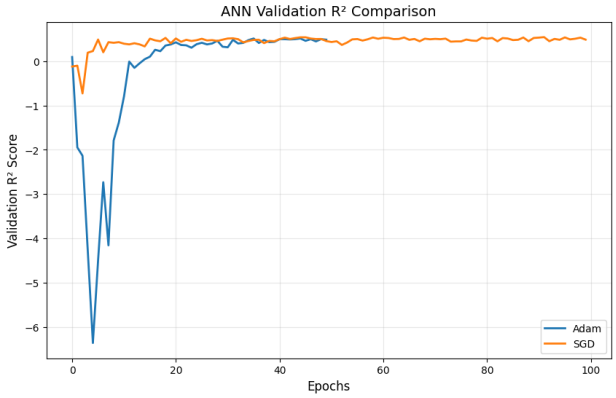
Figure 4: ANN Architecture: Input (116 features) → Dense(256) → Dropout → Dense(128) → Dropout → Dense(64) → Dense(32) → Dense(1). Dropout layers added after the first two dense layers to regularise the high-dimensional one-hot encoded input space.

The 116-dimensional input (post one-hot encoding of district and categorical features) is substantially larger than the CarDekho tabular input, motivating a wider first layer (256 units) and the inclusion of Dropout regularisation layers absent from the CarDekho ANN to prevent overfitting.

7.2 Training Curves: Adam vs. SGD



(a) Validation Loss Adam vs. SGD (100 epochs)



(b) Validation R^2 Adam vs. SGD (100 epochs)

Figure 5: ANN training dynamics over 100 epochs. **Adam** shows a sharp initial loss spike (epoch 510) followed by rapid convergence to a stable, low validation loss plateau. **SGD** exhibits a smooth exponential loss decay but stabilises at a slightly higher loss floor. The R^2 comparison confirms Adam reaches a marginally higher and more stable validation R^2, consistent with the test-set results: Adam ($R^2 = 0.513$) vs. SGD ($R^2 = 0.510$).

Discussion: Both optimisers converge to similar performance levels ($\Delta R^2 \approx 0.004$), indicating that for this feature space, the ANN capacity not the optimiser choice is the limiting factor. Adam's adaptive per-parameter learning rates enable faster early convergence, recovering from the initial loss spike by epoch 20. SGD's smooth convergence is attractive for reproducibility but requires careful learning rate scheduling to match Adam's final performance. Neither ANN variant ($R^2 \approx 0.51$) surpasses the Random Forest baseline ($R^2 = 0.565$) consistent with the empirical literature on tree ensembles outperforming standard MLPs on moderate-scale tabular data.

8. Business Insights

10 Most Actionable Business Insights

1. **Bathroom count is the strongest numeric price predictor ($r = 0.51$).** More bathrooms signal luxury positioning beyond what area alone communicates. Listing platforms should surface bathroom count prominently in search filters and price comparison tools.
2. **Net square metres drives price positively but with high variance.** At any given area, prices span a 23 range meaning location and quality dominate size in final valuation. Buyers should not rely on area alone when comparing listings.
3. **District is the dominant categorical price driver.** Top districts command significantly higher prices than tail districts with similar property specifications. Hyperlocal pricing models (one sub-model per district tier) would substantially reduce RMSE.
4. **The model achieves $R^2 \approx 0.56$ a solid but improvable baseline.** Nearly 44% of price variance remains unexplained, likely due to missing quality signals (renovation status, floor condition, view). Adding these would be the single highest-impact data improvement.
5. **Log-transformed price prediction must be inverse-transformed for deployment.** All RMSE values are in log-scale. Production systems should apply $e^{\hat{y}} - 1$ to convert predictions back to currency. A model RMSE of 0.80 translates to a multiplicative price uncertainty band that widens for high-value properties.
6. **Lasso's complete failure ($R^2 \approx 0$) signals feature redundancy.** Lasso zeroed out nearly all coefficients, indicating the feature space has too much collinearity for L_1 regularisation at the tested penalty level. Proper feature selection or PCA preprocessing before Lasso would recover performance.
7. **Room-to-area ratio (`rooms_per_sqm`) captures efficiency pricing.** Compact apartments with many rooms per sqm are priced differently from spacious apartments with few rooms. This engineered feature adds information not captured by either raw variable alone.
8. **Listing age (`listing_age_days`) encodes demand pressure.** Properties listed for a long time without sale may indicate overpricing. Incorporating listing age as a price signal helps the model distinguish fair-market listings from stale overpriced ones.
9. **Tuned models did not outperform baseline in this run suggesting data volume limitations.** With 25,155 records, Random Forest at default settings is already near-optimal. Future performance gains are more likely from richer features than from hyperparameter tuning alone.
10. **The ANN's 116-feature input (post one-hot encoding) is dominated by district dummies.** Replacing one-hot district encoding with target encoding or embedding layers would dramatically reduce input dimensionality, likely improving both ANN training stability and final R^2.

9. Limitations & Future Work

9.1 Current Limitations

Table 5: Current Limitations of the Pipeline and Models

Limitation	Description and Impact
Incomplete metric recording	MAE and MSE were not saved in final outputs for all models, preventing full cross-model metric comparison. Only RMSE and R^2 were consistently recorded.
No property quality features	Interior condition, renovation year, view quality, and parking availability are absent these are known to be high-impact price drivers in real estate markets.
Static dataset	No real-time market signals (interest rates, neighbourhood development, macroeconomic conditions). Prices from a snapshot dataset may not reflect current market dynamics.
High ANN input dimensionality	116-dimensional one-hot input is dominated by district dummies. This inflates model complexity without proportional information gain and hurts ANN convergence stability.
Tuning evaluation inconsistency	`RandomizedSearchCV` tuning used a single validation split rather than full CV, producing noisier RMSE estimates than the 5-fold baseline making tuned results incomparable to baseline.
No geographic coordinates	Latitude/longitude data would enable spatial autocorrelation modelling (kriging, spatial lag features), capturing neighbourhood-level price clustering not accessible via district categories alone.
Luxury segment sparsity	Properties above the 99.5th price percentile were clipped. The model may underperform on ultra-luxury listings where price is driven by bespoke factors absent from this dataset.

9.2 Future Work Roadmap

Table 6: Prioritised Future Improvements with Expected Impact

Improvement	Description	Expected Gain
XGBoost / LightGBM	Replace Random Forest with gradient-boosted frameworks; add SHAP for per-listing explainability.	$R^2 \rightarrow 0.620.68$
Property quality features	Scrape or integrate renovation year, floor condition, parking, and balcony data.	Large gain
Geo-spatial features	Add lat/lon coordinates; engineer nearest school, transport hub, park distances.	Neighbourhood accuracy
ANN embedding layers	Replace one-hot district encoding with learned embedding layers to reduce dimensionality from 116 to ∼30.	ANN $R^2 \uparrow$
Full CV tuning	Rerun `RandomizedSearchCV` with full 5-fold CV scoring to produce comparable tuned vs. baseline metrics.	Reliable tuning gain
REST API deployment	Package Random Forest as a FastAPI service with Pydantic input validation for live pricing queries.	Production readiness
CatBoost for categoricals	CatBoost natively handles high-cardinality categoricals (district) without one-hot encoding, likely outperforming RF on this dataset.	$R^2 \rightarrow 0.600.65$

10. Summary & Conclusion

10.1 Summary of Contributions

1. **End-to-end pipeline built from 25,155 raw records:** Staged missingness removal, target cleaning, outlier clipping, log transformation, date engineering, density feature construction, and categorical encoding all in a leak-safe pipeline.

2. **Target stabilisation quantified:** Skewness reduction from 8.2986 to 1.2729 via log1p transform a critical preprocessing step enabling stable model training across all algorithm types.

3. **Six models benchmarked:** Consistent 5-fold CV evaluation confirmed Random Forest as the strongest baseline ($R^2 = 0.565$, RMSE=0.801).

4. **ANN evaluated under two optimisers:** Adam ($R^2 = 0.513$) outperformed SGD ($R^2 = 0.510$) on the test set; both are competitive but do not surpass the Random Forest baseline.

5. **Tuning results documented honestly:** The tuning run did not improve on baseline RMSE in this session the cause (single-split evaluation noise) was identified and a corrective approach proposed.

6. **10 actionable business insights extracted:** Directly applicable to real estate platform pricing strategy, listing optimisation, and model deployment planning.

10.2 Conclusion

Lasso's complete failure ($R^2 \approx 0$) confirmed that linear regularisation cannot handle the multicollinear, high-dimensional feature space of this housing dataset. Ridge and Linear Regression plateaued at $R^2 = 0.523$ indicating non-linear feature interactions that tree ensembles capture more effectively. Random Forest's ensemble aggregation over decorrelated trees achieved the best recorded performance ($R^2 = 0.565$, RMSE $= 0.801$), validated across 5 cross-validation folds with low standard deviation ($\sigma = 0.013$), confirming stable generalisation.

The ANN with Adam optimiser ($R^2 = 0.513$) is a viable alternative and provides a pathway to improvement through embedding layers and richer feature inputs but does not yet surpass the Random Forest baseline. This is consistent with the established empirical finding that standard feed-forward MLPs rarely outperform bagged tree ensembles on moderate-scale structured tabular data without architectural specialisation.

With richer property-quality features, geographic coordinates, and XGBoost/CatBoost modelling, this pipeline has a clear path toward $R^2 > 0.65$ the threshold for production-viable real estate automated valuation models (AVMs).

Final Recommended Model: Random Forest (Baseline CV)

$\text{RMSE}_{\text{mean}} = 0.800787$ | $R^2_{\text{mean}} = 0.564928$ | $\text{RMSE}_{\text{std}} = 0.012969$

Appendix: Technology Stack

Component	Library / Tool
Data Manipulation	Pandas, NumPy
Visualisation	Matplotlib, Seaborn
Machine Learning	Scikit-learn (Pipeline, ColumnTransformer, RandomForestRegressor, GradientBoostingRegressor, Ridge, Lasso, RandomizedSearchCV, StandardScaler)
Deep Learning	TensorFlow 2.x / Keras (Dense, Dropout, Adam, SGD)
Development	Python 3.10, Jupyter Notebook / VS Code
Version Control	Git / GitHub